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Chapter 1

Introduction

The rise of decentralized applications (dApps) has revolutionized various sectors by providing transparency, security, and user autonomy. Nevertheless, a multitude of current dApps grapple with hurdles tied to transaction costs, scaling capabilities, and user engagement. IOTA's Tangle, a directed acyclic graph (DAG)-based technology, offers a promising alternative to traditional blockchain systems by eliminating transaction fees and enabling high scalability. This research focuses on leveraging the IOTA Tangle to build fee-free, reward- incentivized dApps, with the added goal of empowering end-users to actively participate in the network as nodes

1.1 Problem Domain

The domain of our project is Distributed Ledger Technology (DLT) concentrating on IOTA Tangle and Decentralized Applications (dApps). It focuses on changing the IOTA Tangle design to integrate a reward-returning architecture for dApps. This project implies the enhancing of a consensus algorithm designed to reward users and their nodes, thereby increasing participation and availability in addition to corresponding structures.

1.2 Problem Statement

The key problem addressed in this research is the need to design and implement an efficient, fee-free, and reward-incentivized framework for dApps on the IOTA Tangle network. In particular, the intention of this endeavor is to:

1.2.1 Incorporate Reward Mechanisms:

Develop and integrate incentive mechanisms that reward users for validating transactions, which can be used to offset transaction costs or enhance user benefits.

1.2.2 Facilitate Node Participation:

Make it feasible and straightforward for dApp users to participate as nodes in the Tangle network, thereby contributing to network security and scalability without requiring significant computational resources or technical expertise.

1.3 Software Requirements Specification (if applicable)

If the research involves the development of software, this section outlines the functional and non-functional requirements of the software to be developed. It details the features, constraints, and performance criteria necessary for the successful implementation of the proposed solution.

Chapter 2

Literature Review

This chapter critically examines existing literature relevant to the research topic. It identifies key studies, theories, and findings, providing a foundation for the proposed research and highlighting its relationship to prior work. This chapter critically examines existing literature relevant to the research topic. It identifies key studies, theories, and findings, providing a foundation for the proposed research and highlighting its relationship to prior work.

2.1 Related Research

2.1.1 Research Item 1

In this paper by Serguei Popov entitled ‘The Tangle,’ a background of IOTA, a cryptocurrency designed for IoT market is provided with explanation of the mathematical concepts at its foundation. The author presents the notion of The Tangle, a new direction in blockchain technologies based on a DAG, more appropriate for implementing machine-to-machine micropayments. Special attention is paid to Markov Chain Monte Carlo (MCMC) algorithm for selection of transaction attachment sites in the Tangle that is more effective and scalable compared to blockchain systems.

2.1.1.1 Summary of the research item

In the paper, the author explores the Tangle of IOTA, a novel distributed ledger system that eliminates mining and transaction fees, making it suitable for microtransactions of IoT environments. In this system, for instance, every new transaction has to vouch for two different previous transactions, thus creating a Directed Acyclic Graph (DAG). This approach precludes some weaknesses seen in the traditional form of a blockchain like

high transaction costs and power vested in the miners. It also deals with different algorithms for transaction approval, for cumulative weight calculation, and with potential attack scenarios, as, for example, the parasite chain attack. This will be followed by the presentation of the MCMC-based tip selection algorithm as a reliable solution to security and network reliability. The paper also describes the vulnerabilities that IOTA was capable of preventing from quantum computing risks in the future.

2.1.1.2 Critical analysis of the research item

The concept presented in "The Tangle," makes a persuasive case for the superiority of IOTA over typical blockchain design for IoT specifically. Launched to overcome one of the biggest shortcomings of blockchain — its inability to support micro transactions where fees can easily eat into the value of the payment — Tangle has been designed to accommodate fee-less transactions. Yet, the mathematical development of its framework is sound, the paper is, however, disappointing lacking real-world use-case scenarios as well as empirical results as to the performance of the Tangle in action to buttress its arguments. On the strategic level, the discussion of threats – from the parasite chain to quantum threats – is indeed exemplary, but the feasibility of the described defenses, especially for large-scale settings, is yet to be proved.

2.1.1.3 Relationship to the proposed research work

The proposed research work on Building a Reward-Incentivize dApps on IOTA's Tangle conforms to the ideas described in the paper "The Tangle." IOTA Tangle analyses both how the IOTA solution proposal can overcome the compared system shortcomings concentrating precisely on a scalability and expense of transactions in the context of the revolutionary blockchain. The absence of the transaction fees in the Tangle platform means that your project will be well placed on this platform since eliminating transaction fees is one way of making dApps more accessible. Moreover, your project initiative of incorporating the reward system in the frequent participation of the users has similarity to the Tangle Consensus which encourages users to validate transactions. Our goal to raise node participation whilst making it easy for users without lots of knowledge of the method is well-supported by the paper's focus on security and size by utilizing MCMCs for tip selection. Therefore, your project extends upon the core values of IOTA's Tangle adding value-related solutions to frequent the network and attract as many users as possible.

2.1.2 Research Item 2

Research paper “Computing Resource Allocation Scheme for DAG-Based IOTA Nodes”, in which the authors examine a solution to improve the response time of the IOTA platform by incorporating load balancing mechanism. IOTA as an ML ledger technology applied to the Internet of Things uses a DAG called the Tangle to solve issues with scalability, cost per transaction, and latency related to the blockchain. In order to solve this problem, this research presents a WLC algorithm for a productive distribution of the computational tasks between the full nodes within the IOTA network.

2.1.2.1 Summary of the research item

The focus of the paper lies in the efficient management of resources for network-external transactions for IOTA’s Tangle, a DAG-based system in which full and light nodes work together to qualify transactions. Light nodes always submit their task requests to full nodes, which have more computational capacity, and thus performance is usually sometimes significantly hampered. To this end, the authors have come up with Weighted Least Connection (WLC) algorithm that determines the distribution of load across the full nodes depending on computational ability of the nodes as well as the number of active connections. Using simulation and experiment the authors show the efficiency of WLC algorithm in allocating the tasks equally to the nodes and making the IOTA network more scalable and efficient. The findings of the study show that there is a reduction of resource exhaustion related problems and enhance system stability of WLC.

2.1.2.2 Critical analysis of the research item

The paper offers an organised approach to solving one of the significant issues in the IOTA Tangle and illustrates how the fair distribution of resources is essential for ensuring the effectiveness of the system. The WLC algorithm adds new knowledge to this field and provides investigators with a potent means to solve load-balance challenges in decentralized IoT settings. However, the various experiments reveal what if cases and still, the paper fails to expound much on challenges that may be characteristic of its real world deployment-o Node availability is dynamic and several external factors for instance, network latency are bound to impact the entire system. Further, although the WLC algorithm enhances the system’s task distributing, the study lacks comprehensive exposition of its behavior under intensely enlarging examples like IoT systems with tens of thousands of nodes. It would be preferred that future work could attempt to address these weaknesses and assess the proposed algorithm in a number of more varied network dimensions.

2.1.2.3 Relationship to the proposed research work

The research paper and our proposed work are connected because the potential improvement to decentralized networks with the help of DAG technology is the central theme of both projects. The paper also provides a solution for the problem of resource allocation for IOTA nodes that would enable a load-balancing of the work endowment to full nodes and light nodes, thus optimizing the task's distribution in order to maintain the IOTA network strong and scalable. This directly supports your work, which is on the use of IOTA's Tangle for constructing fee less, rewards-based dApps . In terms of its focus, which is to introduce reward mechanisms that will motivate the users of nodes as well as increase activity in the system, both works respond to some of the main objectives essential for enhancing the efficiency of decentralized systems based on DAG structures.

2.1.3 Research Item 3

This research paper lies in its contribution to the security of decentralized ledger technologies, particularly IOTA. By providing a framework for detecting Parasite Chain attacks, the paper enhances the resilience of the Tangle against adversarial actions. The proposed detection mechanisms not only improve the integrity of the IOTA network but also offer insights into the broader implications for other DAG-based systems facing similar security challenges. This work underscores the importance of proactive measures in maintaining the trust and reliability of decentralized financial systems

2.1.3.1 Summary of the research item

The paper focuses on the detection of a specific type of attack known as the Parasite Chain attack within the IOTA cryptocurrency, which utilizes a directed acyclic graph (DAG) structure called the Tangle. The authors propose a detection mechanism that leverages the distribution of transaction approvers to identify anomalies indicative of such attacks. By modeling the expected distribution of approvers in a non-malicious environment, the paper introduces metrics to compare this reference distribution with actual measurements, allowing for the detection of malicious behavior. The study emphasizes the importance of sampling methods and presents two different approaches to enhance detection efficacy

2.1.3.2 Critical analysis of the research item

The research evolves from the understanding of how decentralized systems like IOTA can be vulnerable to specific attacks, particularly those that manipulate the consensus mechanism. The authors critically analyze existing tip selection algorithms and their susceptibility to adversarial actions. They introduce analytical models for both Uniform Random

Tip Selection (URTS) and Unbiased Random Walk (URW), demonstrating their interrelation and the impact of transaction sampling on the likelihood of successful attacks. This evolution highlights the need for robust detection mechanisms in the face of increasingly sophisticated threats to decentralized networks

2.1.3.3 Relationship to the proposed research work

Our work and the paper aim at IOTA's tree called Tangle and tackle the primary inefficiencies in distributed systems. Whereas, your work is largely concentrated on creating more decentralized applications that do not attract any fees and also uses the reward system to attract more users, the other paper is protection of the Tangle to prevent some attacks such as the Parasite Chain that tamper with the ledger. Both methods essentially have the same purpose of creating a better Tangle network with added security and encouraged more stakeholder engagement.

2.1.4 Research Item 4

This paper is limited in its focus only to the platform's nodes. It may be said that the primary purpose of the reward is encourage truthful behavior among the participants, to build up a stable, safe and highly effective decentralized network. In the proposed system there are incentives to help participants manage themselves and dictating what they are supposed to do. Though this paper provides an incentive system specifically for Waterfall, the ideas discussed here can be generalized to other Proof-of-Stake (PoS) consensus mechanisms in decentralized systems. It gives an important background on fair reward allocation strategies for the platform workers and the appropriate punishments for a mischievous action. Token omics is a term derived from the grapple with the economics of tokens and is the basis for this system. Of particular importance for rewarding the desired behavior, which is for example, processing transactions and validation of the blocks is the incentive mechanism. But it also has to be coupled with proper punishments for vectors in the network to deter malicious actions, to keep the network integrity.

2.1.4.1 Summary of the research item

The subject of the study is the development of the motivational model for decentralized platforms from the DA* technology group. It recommends measures on how compensation for the effort made by the participants known as Workers will be provided for while discouraging antisocial activities. The objective is to keep a decentralized network with good consensus algorithm, sustainable and reliable against fundamental adversities such as Sybil, Splitting and Nothing-at-Stake attack. The platform employed in this study is

called Waterfall –, that currently and in the future has the goal of coming up with appropriate token omics, while being able to scale, decentralize, and incentivize honest actors economically.

2.1.4.2 Critical analysis of the research item

The incentive value as proposed for the project shall aim at fostering participation of the users in the network while at the same time ensuring the network cannot be vulnerable to these attacks. It underlines establishing a structure in which the consequences of improper actions are worse than the benefits, which means that destructive actions become counterproductive. However, its performance in real-world decentralized environments which are inherently more complex, random and dynamic have not been well explored. The first advantage of the study is the multi-parameter nature of the project – the optimization of the network, for both performance, security, and reliability. However, a critical problem is in how to effectively manage and model, in a largely decentralized environment, the real-time malicious actions, something that will need considerable more future work and modeling. Furthermore, the reward and penalty system has been constructed theoretically, yet there are some questions rise about its applicability in real networks, particularly in a large scale networks. In general, we have a well-designed solution that can be considered to be quite basic for the system under consideration; however, a real-world testing of the proposed system can undoubtedly imagen-enhance its performance in a variety of situations.

2.1.4.3 Relationship to the proposed research work

The proposed research will have a strong correlation with the incentive system discussed in the paper referred above. Both works are devoted to distributed platforms based on DAG structures: your study of Tangle in IOTA and the paper you mentioned on Waterfall. As your research goals are to create decentralized applications with no transaction fees and to use people's engagement as nodes in the network, the paper also focuses on encouraging honest miners and punishing cheaters in order to protect the network. In regard to the weaknesses of the DAG-based systems, both approaches consider the issues related to scalability, the user engagement level, and transaction costs, whereas your work focuses on removing the transaction fees and increasing the users' activity.

2.1.5 Research Item 5

The paper Scalable Off-Chain Instant Payments authored by Joseph Poon and Thaddeus Dryja presents a fix to the scalability challenge on the Bitcoin blockchain. The key sub-

ject under consideration is the indisputable fact that bitcoin as it is does not have sufficient capacity to stimulate and perform global financial transactions per se, based on its technological devices, namely the block-chain technology. The authors present the off-chain micropayment channel solution called the Lightning Network as one capable of facilitating faster, cheaper and more scalable transactions while retaining the decentralized nature of Bitcoin. The paper also describes how these channels operate using time-locked contracts as well as cryptographic validation for dependable and trustless bilateral transactions offline.

2.1.5.1 Summary of the research item

The paper goes further to identify the scalability as the challenge with the current model of blockchain whereby the Bitcoin blockchain can only handle 7 transactions per second compared to Visa's 47, 000. To address this, the authors introduce the Lightning Network as an off-chain layered system of bidirectional micro payment channels where multiple transactions are possible between contracting parties without burdening the block chain. Such off-chain transactions occur when the need arises and when such transactions are put on the blockchain, they employ features such as time lock and multiple signatures. Moreover, utilization of Hashed Time lock Contracts (HTLCs) that underpin the Lightning Network mean that multi-hop payments are also possible and with scalability to billions of transactions per day for micro payments and yet virtually no fee.

2.1.5.2 Critical analysis of the research item

The Lightning Network is a major enhancement of Bitcoin scalability, which presents an off-chain payment solution for Bitcoin that is decentralized, instantaneous, and secure while avoiding the burden of bloating the blockchain with micropayment data. It is most appropriate for micropayment since it has a feature of micropayment channels, which solve the problem of high costs and time consumption in BTC. Nevertheless, the following key issues are still questionable: Channel capacity problem bottlenecks which may hinder users to accommodate for higher value payments and the problem of securing and increasing the strength of the participants' network for relaying the payments. Furthermore, the supposed taxing of few channels, and many transactions might demotivate average users; the supposed future usage of a Bitcoin protocol soft-fork to deal with malleability issues might also be a problem since the community might not like it.

2.1.5.3 Relationship to the proposed research work

The Lightning Network and IOTA's Tangle are two of the most promising methodologies trying to improve the performance and functionality of the distributed ledger technolo-

gies by cutting down the on-chain transaction and ensuring high scalability. While the LN wants to vastly reduce fees for micropayment channels, your project utilizes IOTA's feeless Tangle to incentivize participants, employing a reward system for validating those transactions. As we will compare later, both strategies aim at increasing user engagement and lowering the cost barrier of dApps.

2.2 Analysis Summary of research items

Research items	Critical analyses
The Tangle	The concept presented in "The Tangle," makes a persuasive case for the superiority of IOTA over typical blockchain design for IoT specifically. Launched to overcome one of the biggest shortcomings of blockchain — its inability to support micro transactions where fees can easily eat into the value of the payment — Tangle has been designed to accommodate fee-less transactions. Yet, the mathematical development of its framework is sound, the paper is, however, disappointing lacking real-world use-case scenarios as well as empirical results as to the performance of the Tangle in action to buttress its arguments. On the strategic level, the discussion of threats – from the parasite chain to quantum threats – is indeed exemplary, but the feasibility of the described defenses, especially for large-scale settings, is yet to be proved.

Research paper “Computing Resource Allocation Scheme for DAG-Based IOTA Nodes”,	The paper offers an organised approach to solving one of the significant issues in the IOTA Tangle and illustrates how the fair distribution of resources is essential for ensuring the effectiveness of the system. The WLC algorithm adds new knowledge to this field and provides investigators with a potent means to solve load-balance challenges in decentralized IoT settings. However, the various experiments reveal what if cases and still, the paper fails to expound much on challenges that may be characteristic of its real world deployment-o Node availability is dynamic and several external factors for instance, network latency are bound to impact the entire system. Further, although the WLC algorithm enhances the system’s task distributing, the study lacks comprehensive exposition of its behavior under intensely enlarging examples like IoT systems with tens of thousands of nodes. It would be preferred that future work could attempt to address these weaknesses and assess the proposed algorithm in a number of more varied network dimensions.
Research paper Parasite Chain attacks	The research evolves from the understanding of how decentralized systems like IOTA can be vulnerable to specific attacks, particularly those that manipulate the consensus mechanism. The authors critically analyze existing tip selection algorithms and their susceptibility to adversarial actions. They introduce analytical models for both Uniform Random 6 2.1 Related Research Tip Selection (URTS) and Unbiased Random Walk (URW), demonstrating their interrelation and the impact of transaction sampling on the likelihood of successful attacks. This evolution highlights the need for robust detection mechanisms in the face of increasingly sophisticated threats to decentralized networks

Research paper decentralizes platforms by DA technology	The incentive value as proposed for the project shall aim at fostering participation of the users in the network while at the same time ensuring the network cannot be vulnerable to these attacks. It underlines establishing a structure in which the consequences of improper actions are worse than the benefits, which means that destructive actions become counterproductive. However, its performance in real-world decentralized environments which are inherently more complex, random and dynamic have not been well explored. The first advantage of the study is the multi-parameter nature of the project – the optimization of the network, for both performance, security, and reliability. However, a critical problem is in how to effectively manage and model, in a largely decentralized environment, the real-time malicious actions, something that will need considerable more future work and modeling. Furthermore, the reward and penalty system has been constructed theoretically, yet there are some questions rise about its applicability in real networks, particularly in a large scale networks. In general, we have a well-designed solution that can be considered to be quite basic for the system under consideration; however, a real-world testing of the proposed system can undoubtedly image-enhance its performance in a variety of situations.
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The paper Scalable Off-Chain Instant Payments authored by Joseph Poon and Thaddeus	The Lightning Network is a major enhancement of Bitcoin scalability, which presents an off-chain payment solution for Bitcoin that is decentralized, instantaneous, and secure while avoiding the burden of bloating the blockchain with micropayment data. It is most appropriate for micropayment since it has a feature of micropayment channels, which solve the problem of high costs and time consumption in BTC. Nevertheless, the following key issues are still questionable: Channel capacity problem bottlenecks which may hinder users to accommodate for higher value payments and the problem of securing and increasing the strength of the participants' network for relaying the payments. Furthermore, the supposed taxing of few channels, and many transactions might demotivate average users; the supposed future usage of a Bitcoin protocol soft-fork to deal with malleability issues might also be a problem since the community might not like it.
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Chapter 3

Proposed Approach

To propose a solution for the research problem, which is to design a fee-free, reward-incentivized framework for decentralized applications (dApps) using IOTA's Tangle, we can break down the approach into different stages. Below is a methodology and theoretical framework, including a Consensus Algorithm and Reward Mechanism that addresses the research problem.

3.1 Theoretical Framework

The research focuses on developing a modified consensus protocol for IOTA's Tangle, integrating a reward mechanism to incentivize users and ensure greater participation as nodes. The approach is twofold:

3.1.1 Consensus Protocol Modification:

- Adapt the existing IOTA Tangle to include lightweight nodes with low computational overhead.
- Introduce a dynamic reward system that compensates participants for validating transactions.
- The reward mechanism will be designed to operate seamlessly with the IOTA Tangle. It aims to encourage participation by providing incentives for validating transactions and contributing to the network's security and scalability.

3.2 Methodology

The methodology involves several stages:

3.2.1 Research and Literature Review

- Analyze current consensus protocols in Distributed Ledger Technologies (DLTs), with a specific focus on IOTA and Directed Acyclic Graph (DAG)-based consensus mechanisms.
- Investigate reward systems used in decentralized applications to explore efficient, scalable models that encourage participation.

3.2.2 Design of Reward Incentive Mechanism

- Develop a reward structure that fairly compensates users for validating transactions and reduces transaction costs.
- Define how rewards can be earned and spent, ensuring that the reward mechanism integrates well with the IOTA ecosystem.

3.2.3 Node Participation Enhancement

- Create interfaces and lightweight node implementations to make it easier for users to act as nodes.
- Ensure that users with limited resources (e.g., low computational power) can still contribute meaningfully to the network.

3.2.4 Integration with IOTA Tangle

- Extend IOTA's existing Tangle protocol to include the reward mechanism.
- Ensure the modifications maintain IOTA's scalability, security, and fee-free nature.

3.2.5 Simulation and Testing

- Use simulation tools like Hyperledger Fabric to test the performance of the proposed system under various network conditions.

- Analyze metrics such as scalability, latency, network resilience, and security.

3.3 Consensus Algorithm Framework

The consensus algorithm framework will include the following components:

3.3.1 Node Validation Mechanism:

Users participate as nodes by validating transactions and each transaction must approve two previous transactions in the Tangle, creating a DAG structure.

3.3.2 Reward System:

Nodes that validate transactions receive rewards based on their contributions and rewards are generated dynamically based on the network's activity and the amount of computational power contributed by the node.

3.3.3 Lightweight Node Implementation:

Enable resource-constrained users to participate as nodes by minimizing the computational requirements and provide a simplified interface for these users to validate transactions.

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